

AI Operations Associate

Take-Home Task · Screening Stage

Time to spend	4–6 hours of focused work
Window	5 calendar days from receipt
Submit	One PDF write-up (max 3 pages) + your working AI prototype + prompt history

This is a screening task. We are not looking for a polished consulting deck or a finished product. We want to see how you think: how you make sense of a messy situation, how you decide what actually matters, and how you use AI to do something useful about it. A rough prototype that solves the right problem beats a slick one that solves the wrong problem.

The situation

BharatTrip is a B2B travel company. It sells flights, hotels and packages to travel agents, who in turn serve travellers. You do not need to know anything about the travel industry to do this task — everything you need is in the brief and the attached data.

When an agent cancels a booking, they ask BharatTrip for a refund. Two teams handle every refund:

- The **Support team** talks to agents, takes the refund request, and tracks it in a spreadsheet.
- The **Finance team** checks the request, pays the money out, and tracks it in a *separate* spreadsheet.

Each team built its own tracker over time, in its own way. On top of these two sheets, some refund requests arrive informally — by email or messaging app — and are handled off-tracker.

Leadership's problem: customer and agent escalations about refunds have been climbing for two months. Refund *volume* has stayed roughly flat over the same period. Nobody can say clearly why the complaints are rising, and different people give different explanations. Leadership wants the escalations brought down — **without adding headcount**.

What you have

One Excel file, **BharatTrip_Refund_Data.xlsx**, with four tabs:

- **Read_Me** — a plain-English description of the data (start here).
- **Support_Tracker** — the Support team's record of refund requests.
- **Finance_Tracker** — the Finance team's record of refund payouts.
- **Escalations** — recent complaints that were raised, and how long each has been open.

The two trackers are *supposed* to describe the same refunds, but they were built independently. Part of the work is figuring out how they relate to each other and where they disagree. The data is synthetic and anonymised, but shaped like the real thing — messy in the ways real operational data is messy.

A few requests that never made it into a tracker

These came in informally and were never logged. They are included here exactly as they arrived. Treat them as part of the picture:

Email — 5 June, from Peak Journeys:

"Hi, I cancelled booking for DEL-DXB last week and was told I'd get a refund. I haven't received anything and I don't have any reference number. Please help, my client is asking. (This is the case later referenced as RF-1099.)"

WhatsApp — 8 June, from Nomad Travel:

"Bhai refund ka kya hua? BLR-MAA wala. 2 hafte ho gaye, koi jawab nahi. (Later referenced as RF-1098.)"

Email — 11 June, internal note from a Support agent:

"Finance says they never received several of the refunds I closed last month. I marked them done on my side. Not sure where they went."

What we'd like you to do

Work through the following. There is no single correct answer — your reasoning is what we're assessing.

1. Define the problem.

In your own words, what is actually going wrong here? Not the symptom ("escalations are up") but your best diagnosis of the underlying cause(s). Use the data to back it up.

2. Analyse.

Dig into the data and show what you found. What patterns explain the escalations? Where do the two trackers agree, disagree, or fall silent? What would you want to validate if you had more time or access?

3. Design a solution.

Propose how BharatTrip should run this process so escalations come down and stay down — given the no-new-headcount constraint. Be explicit about trade-offs: what you would automate, what you would deliberately keep manual, and why. What is the single source of truth, and how do the two teams stay in sync?

4. Build an AI prototype.

Build something working that uses AI to solve a *meaningful slice* of the problem. It does not have to be the whole solution. It can be an agent, a workflow, an automation, a script, a chatbot, a reconciliation tool — whatever you can justify. Use whatever tools you like (no-code, code, off-the-shelf AI, or a mix). We care that it runs and that the choice of what to build shows judgement.

5. Measure.

How would you know your solution worked? Name the one or two metrics you'd watch, and what a good result looks like after, say, a month.

What to submit

- A **write-up of max 3 pages** (PDF) covering points 1, 2, 3 and 5 above. Concise is good. We prefer clear thinking over length.
- Your **working prototype** — a link (hosted app, Colab, repo, video of it running) or files. Include a one-line note on how to run or view it.
- Your **prompt history** — the actual prompts you used with AI tools while doing this task. Copy-paste or export is fine. We want to see how you work with AI, not just the output.

A few notes

- **Use AI throughout.** This role is about solving operational problems with AI as your default tool. Show us that.
- **Write down your assumptions.** The brief is deliberately incomplete, exactly like real work. If something is unclear, make a reasonable call and note it — don't wait to be told.

- **You won't get more information.** Everything you need is here. If you feel a question is genuinely blocking, state the assumption you made to move past it.
- **Quality of thinking > quality of presentation.** A plain document that reasons well will score higher than a beautiful one that doesn't.

Questions that genuinely block you? Reply to the email you received this from, with the specific question and the assumption you'll proceed on if we don't answer in time. We read these — how you ask is itself a signal.